FUTURE SHOCK INDEX

رواد مصر الرقمية

GLOBAL RESILIENCE

DEPI ROUND 4 - YAT MAADI

Who survives the shock?

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MEET THE TEAM

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Dr. Amal Mahmoud | Supervisor

Eslam Hassan

Habiba Ahmed

Mostafa Elrkhawy

Sara Mohamed

Omar Mahmoud

Ann Osama

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AGENDA

Introduction
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Objectives
02

Methodology
03

Insights
04

Recommendations
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INTRODUCTION

Measuring how prepared countries are for future global shocks

	100 Countries
	2000–2023
	14 Indicators
	7 Domains
	14 World Bank + FAO Data

A comprehensive resilience assessment that identifies countries' strengths, vulnerabilities, and long-term trends.

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PROJECT OBJECTIVE

- Build a Global Resilience Index for 100 countries (2000–2023)
- Design a Galaxy Schema with 2 fact tables and 4 dimension tables
- Perform statistical analysis of trends and outliers

- Build an end-to-end analytics solution across five tools
- Normalize and align all indicators
- Deliver actionable policy recommendations

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METHODOLOGY

Data Collection
Integrate 15 datasets covering 100 countries and 14 indicators.

Data Cleaning
Correct data types and exclude 2024-2025 data.

Data Modeling
Build a Galaxy Schema with 2 facts and 4 dimensions

Normalize & Invert
Standardize all indicators to a common scale.

Analysis & Dashboards
Build analytical queries, dashboards, and calculated measures.

ABOUT THE DATASET

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- Access to electricity.xlsx
- Clean Fuel Access.xlsx
- Electricity consumption .xlsx
- FAO Food Price Index.xlsx
- Fixed broadband subscriptions.xlsx
- Food Imports _ of merchandise imports.x...
- GDP Growth.xlsx
- Global Health Expenditure.xlsx
- Hospitals.xlsx
- Inflation.xlsx
- Internet Users.xlsx
- Physicians.xlsx
- Political Stability.xlsx
- Prevalence of Undernourishment.xlsx
- Renewable energy .xlsx

Converted 15 datasets from wide to long format for analysis.

	A	B	C	D	E	F	G	H	I	J	
1	Data Source	World Development Indicators									
2	Last Updated Date	24/02/2026									
3											
4	Country Name	Country Code	Indicator Name	Indicator Code	1960	1961	1962	1963	1964	1965	1966
5	Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
6	Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
7	Afghanistan	AFG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
8	Africa Western and Central	AFW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
9	Angola	AGO	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
10	Albania	ALB	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
11	Andorra	AND	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
12	Arab World	ARB	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
13	United Arab Emirates	ARE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
14	Argentina	ARG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
15	Armenia	ARM	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
16	American Samoa	ASM	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
17	Antigua and Barbuda	ATG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
18	Australia	AUS	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	3.728813559	2.287581699	-0.31948882	0.641025641	2.866242038	3.405572755	3.29
19	Austria	AUT	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	1.945749352	3.542239867	4.381798639	2.708766995	3.868563549	4.930916377	2.05
20	Azerbaijan	AZE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
21	Burundi	BDI	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							4.43
22	Belgium	BEL	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	0.299467255	0.992676261	1.404607035	2.148003413	4.168760586	4.065201375	4.17
23	Benin	BEN	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
24	Burkina Faso	BFA	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	7.788161994	18.56213873	1.675909561	5.574202829	1.845123198	-0.73025252	2.3
25	Bangladesh	BGD	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
26	Bulgaria	BGR	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
27	Bahrain	BHR	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							2.13
28	Bahamas, The	BHS	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
29	Bosnia and Herzegovina	BIH	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
30	Belarus	BLR	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
31	Belize	BLZ	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
32	Bermuda	BMU	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
33	Bolivia	BOL	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	11.53315258	7.561493072	5.879331422	-0.70635721	10.18181818	2.860286029	6.95
34	Brazil	BRA	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
35	Barbados	BRB	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
36	Brunei Darussalam	BRN	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
37	Bhutan	BTN	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
38	Botswana	BWA	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
39	Central African Republic	CAF	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
40	Canada	CAN	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	1.358695652	1.018766756	1.061571125	1.628151261	1.912144703	2.332657201	3.81
41	Central Europe and the Baltics	CEB	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							

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UNPIVOTING THE DATASETS

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Country Name	Country Code	Indicator Name	Indicator Code	Year	Value
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2000	4.044021312
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2001	2.883604303
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2002	3.315246786
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2003	3.656365079
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2004	2.529129469
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2005	3.397786778
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2006	3.608024388
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2007	5.392567845
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2008	8.955987052
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2009	-2.135428721
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2010	2.078140719
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2011	4.316297422
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2012	0.627471993
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2013	-2.372065243
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2014	0.421440919
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2015	0.474763597
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2016	-0.931195987
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2017	-1.028281788
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2018	3.626041414
Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2019	4.257462043
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2000	8.601485149
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2001	5.840353983
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2002	8.763755203
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2003	7.44970014
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2004	5.023420686
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2005	8.558038022
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2006	8.898163802
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2007	8.450774806
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2008	12.56664466
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2009	8.954218024
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2010	5.537537917
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2011	8.97120641
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2012	9.158708432
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2013	5.748831336
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2014	5.525819234
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2015	5.098561644
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2016	6.446877223
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2017	6.221375289
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2018	4.689806468
Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	2019	4.102850755

DATASET DESCRIPTION

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	A	B	C	D	E	F	G
1	Indicator Code	Country Code	Country_Key	Indicator_Key	Year_Key	Value	Normalized Value
2	IT.NET.BBND.P2	ALB	1	1	5	0.0088422	0.01
3	IT.NET.BBND.P2	ALB	1	1	7	0.33081	0.02
4	IT.NET.BBND.P2	ALB	1	1	8	2.13837	0.05
5	IT.NET.BBND.P2	ALB	1	1	9	3.10656	0.07
6	IT.NET.BBND.P2	ALB	1	1	10	3.60359	0.08
7	IT.NET.BBND.P2	ALB	1	1	11	4.40357	0.10
8	IT.NET.BBND.P2	ALB	1	1	12	5.5013	0.12
9	IT.NET.BBND.P2	ALB	1	1	13	6.27864	0.14
10	IT.NET.BBND.P2	ALB	1	1	14	7.16191	0.16
11	IT.NET.BBND.P2	ALB	1	1	15	8.37878	0.18
12	IT.NET.BBND.P2	ALB	1	1	16	9.19224	0.20
13	IT.NET.BBND.P2	ALB	1	1	17	10.4757	0.22
14	IT.NET.BBND.P2	ALB	1	1	18	12.5058	0.26
15	IT.NET.BBND.P2	ALB	1	1	19	15.1193	0.32
16	IT.NET.BBND.P2	ALB	1	1	20	17.7209	0.37
17	IT.NET.BBND.P2	ALB	1	1	21	19.6304	0.41
18	IT.NET.BBND.P2	ALB	1	1	22	20.6989	0.43
19	IT.NET.BBND.P2	ALB	1	1	23	22.5049	0.47
20	IT.NET.BBND.P2	ARE	2	1	5	2.77226	0.07
21	IT.NET.BBND.P2	ARE	2	1	0	0.0564751	0.01
22	IT.NET.BBND.P2	ARE	2	1	1	0.22489	0.01
23	IT.NET.BBND.P2	ARE	2	1	2	0.420337	0.02
24	IT.NET.BBND.P2	ARE	2	1	3	0.723371	0.02
25	IT.NET.BBND.P2	ARE	2	1	4	1.2677	0.04
26	IT.NET.BBND.P2	ARE	2	1	6	4.80366	0.11
27	IT.NET.BBND.P2	ARE	2	1	7	6.75244	0.15

Fact table: Global_Fact_Indicators
Total Records: 31,899 rows
Total Columns: 7 columns

- Unpivoted 15 Excel datasets into long format.
- Appended all datasets into a single Fact_Global_Indicators table.
- Standardized data for analysis across 100 countries (2000–2023).

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DATASET DESCRIPTION

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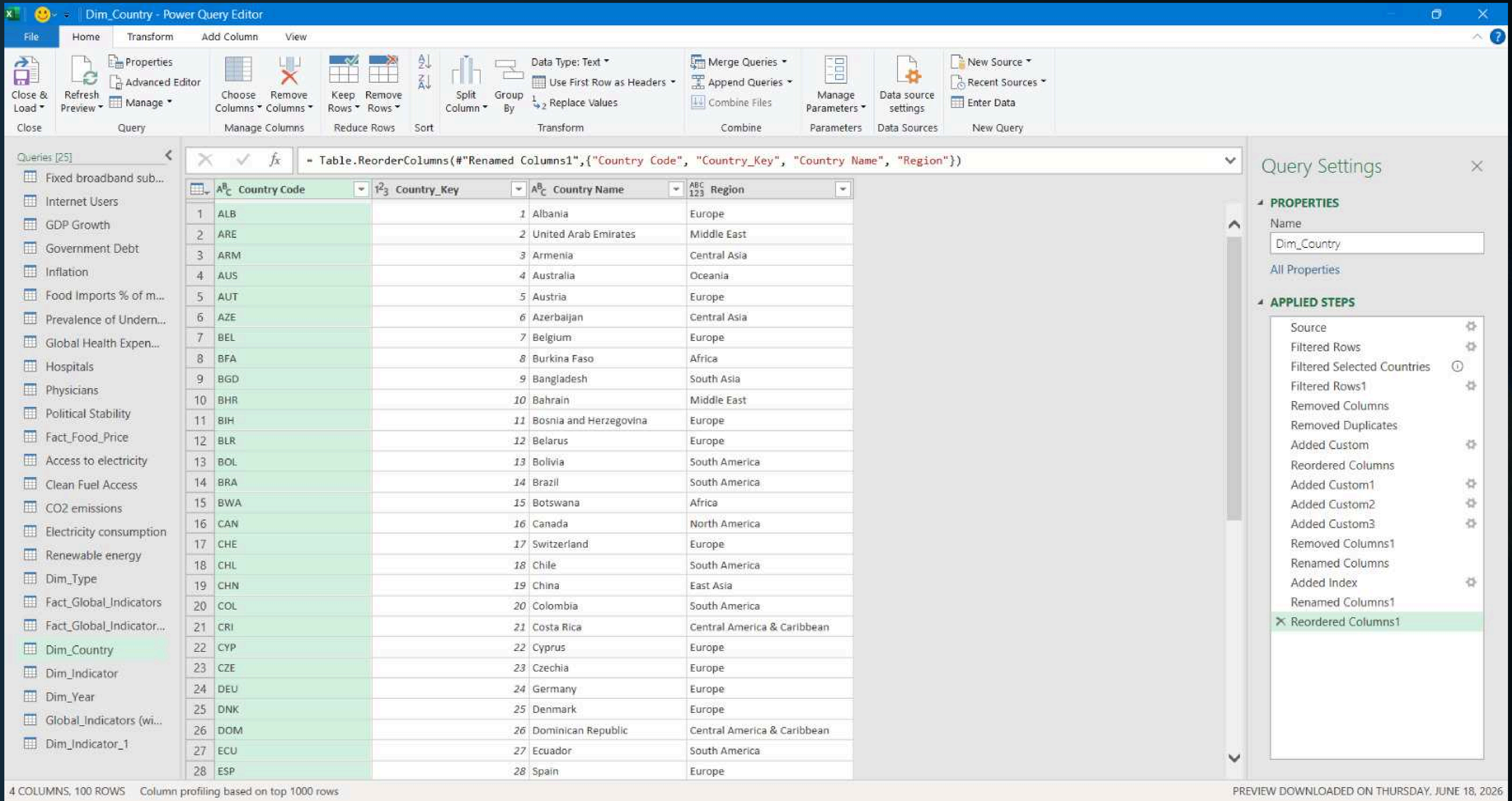
Fact table: Fact_Food_Index

Year_Key	Type_Key	Month	Price Index	Price Category
0	1	1	66.45472637	Cheap
0	1	2	66.0217187	Cheap
0	2	1	76.17888612	Slightly Cheap
0	3	1	63.07155132	Cheap
0	4	1	65.55746983	Cheap
0	5	1	61.09502841	Cheap
0	6	1	43.67388534	Extremely Cheap
0	2	2	76.53660046	Slightly Cheap
0	3	2	63.43071526	Cheap
0	4	2	65.6925333	Cheap
0	5	2	58.30988886	Very Cheap
0	6	2	41.16870532	Extremely Cheap
0	1	3	66.79530545	Cheap
0	2	3	78.95359848	Slightly Cheap
0	3	3	62.94856085	Cheap
0	4	3	65.05010705	Cheap
0	5	3	60.13416887	Cheap
0	6	3	40.00135073	Extremely Cheap
0	1	4	67.52972893	Cheap
0	2	4	78.35434279	Slightly Cheap
0	3	4	63.39696572	Cheap
0	4	4	65.16861367	Cheap
0	5	4	62.24640238	Cheap
0	6	4	46.84983101	Extremely Cheap
0	1	5	67.86517904	Cheap

PREPROCESSING USING EXCEL

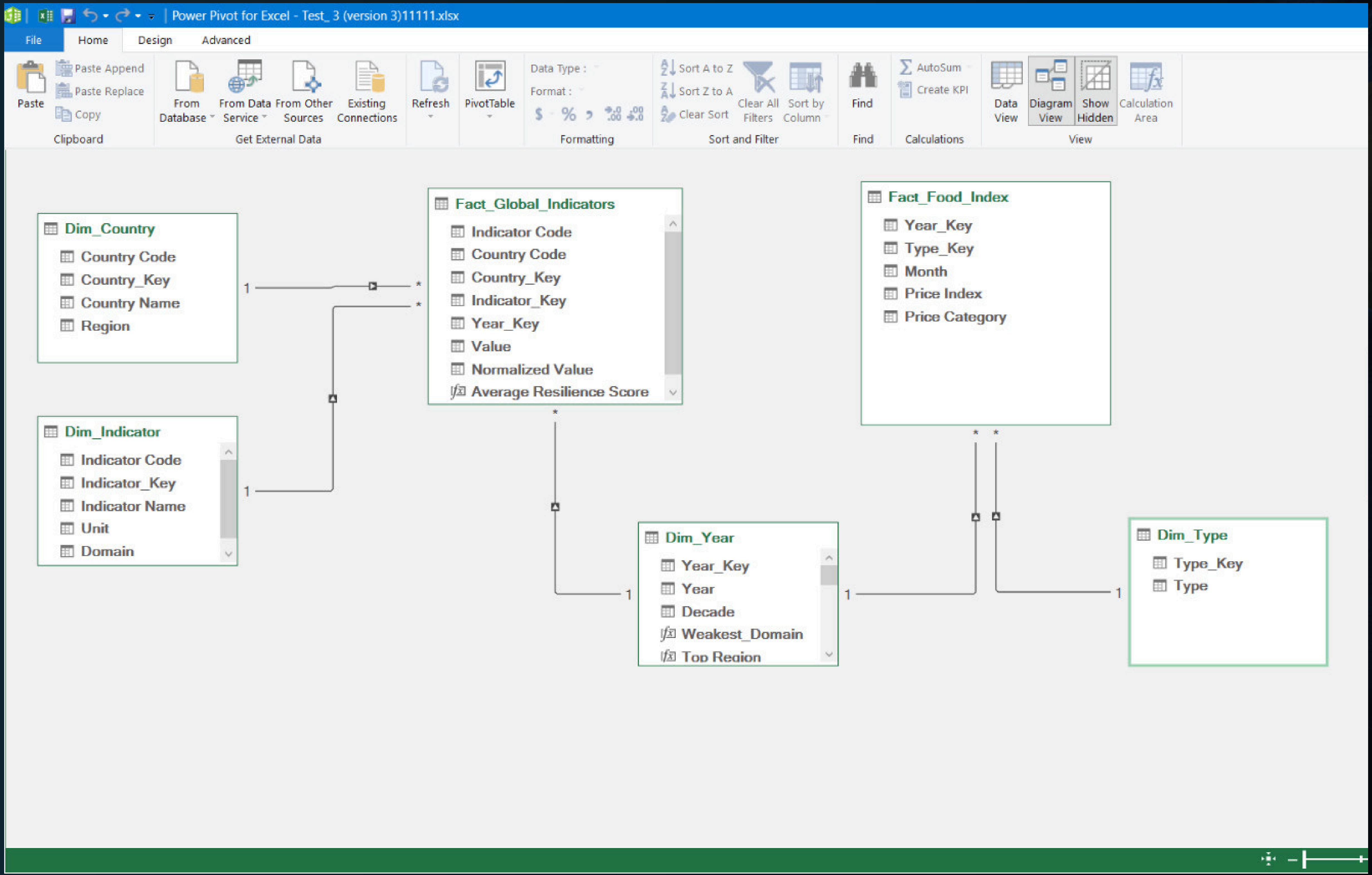


- **Performed data cleaning and transformation in Power Query.**
- **Built a galxey schema with 2 fact tables and 4 dimension tables.**
- **Prepared Dim_Country by filtering 100 countries.**

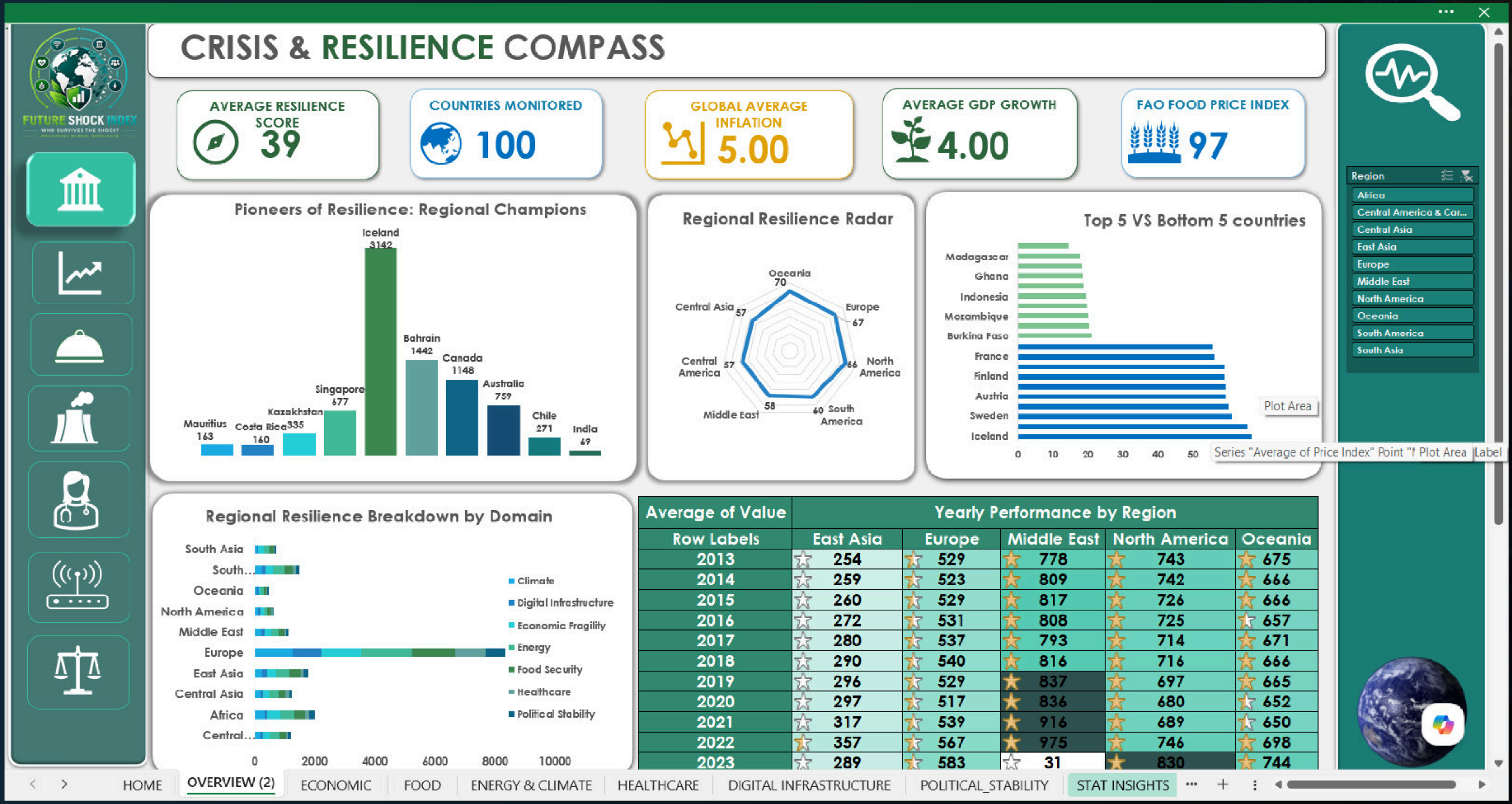
Created the Dim Type dimension table

- **Mapped six food categories using unique type keys.**
- **Ensured consistent food classification throughout the analysis.**

DATA MODELING IN EXCEL



DASHBOARD USING EXCEL



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DASHBOARD USING EXCEL

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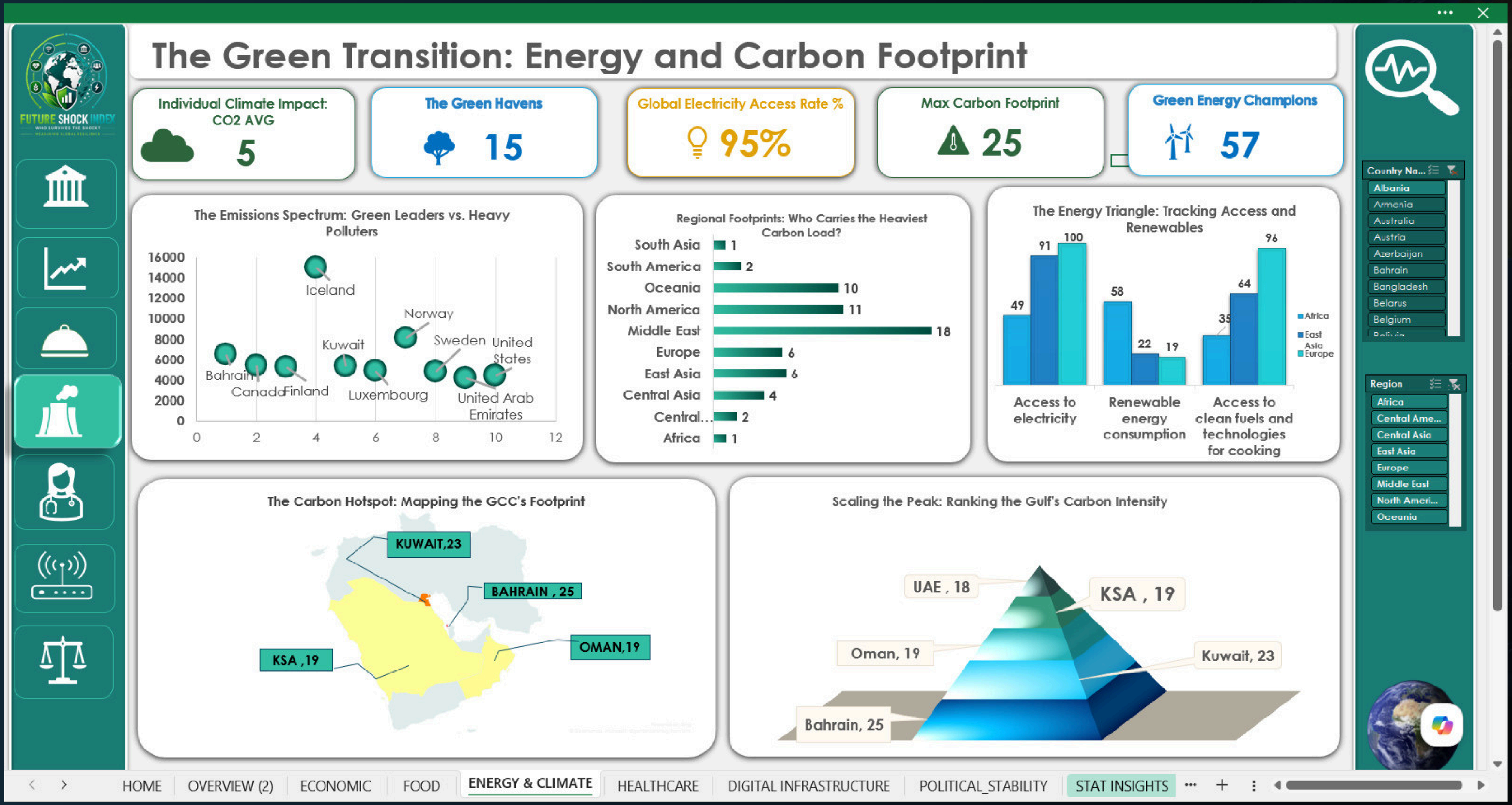
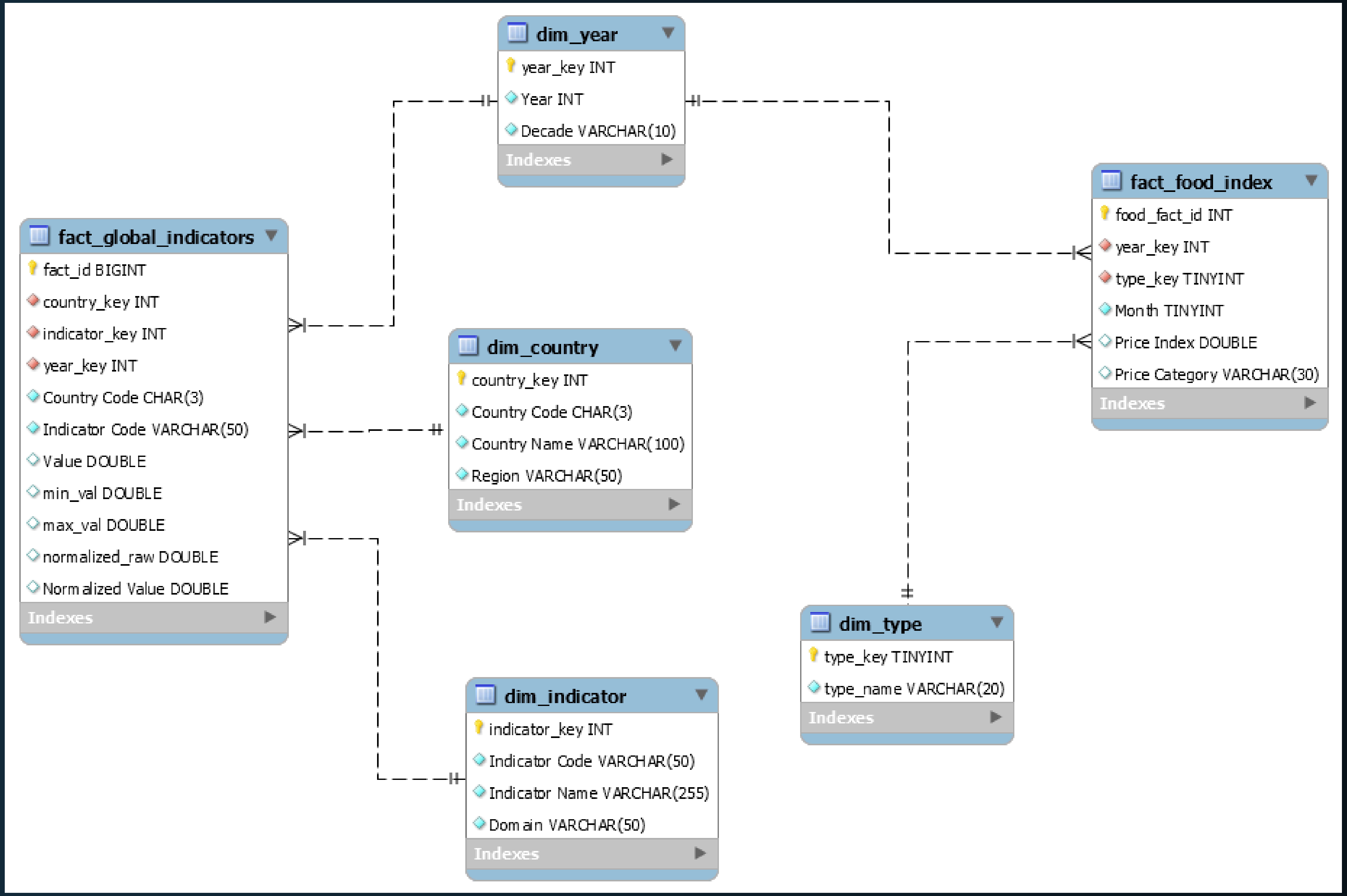


DATA MODELING USING SQL



Schema Architecture

- Integrated 14 global datasets into a centralized data warehouse.
- Designed a Galaxy Schema with Fact and Dimension tables.
- Created a scalable foundation for analytics and KPI generation.

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HIGH-RISK COUNTRIES

MySQL

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Country Name	Region	Composite Score	Risk Category
Madagascar	Africa	0.36	High Risk
Rwanda	Africa	0.36	High Risk
Bangladesh	South Asia	0.37	High Risk
Mozambique	Africa	0.37	High Risk
Ethiopia	Africa	0.38	High Risk
Burkina Faso	Africa	0.39	High Risk
Togo	Africa	0.39	High Risk
Pakistan	South Asia	0.41	High Risk
Zambia	Africa	0.41	High Risk
Ghana	Africa	0.43	High Risk
Nepal	South Asia	0.44	High Risk
Philippines	East Asia	0.44	High Risk
Botswana	Africa	0.45	High Risk
Indonesia	East Asia	0.45	High Risk
Sri Lanka	South Asia	0.45	High Risk
Nicaragua	Central A...	0.45	High Risk
India	South Asia	0.46	High Risk
Cambodia	East Asia	0.46	High Risk
Tajikistan	Central Asia	0.46	High Risk

Result 25

The table lists countries categorized as High Risk according to their overall Composite Score.

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DOMAIN SCORES

MySQL

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The table provides a breakdown of each country's resilience across the six domains.

Country Name	Region	Climate & Energy	Digital Infrastructure	Economic Fragility	Food Security	Healthcare	Political Stability	Composite Score
Cambodia	East Asia	0.47	0.1	0.67	0.79	0.14	0.57	0.46
Tajikistan	Central Asia	0.64	0.1	0.6	0.47	0.29	0.42	0.46
India	South Asia	0.54	0.08	0.68	0.8	0.12	0.39	0.46
Nicaragua	Central Ameri...	0.56	0.12	0.7	0.55	0.19	0.55	0.45
Botswana	Africa	0.49	0.19	0.7	0.56	0.17	0.84	0.45
Sri Lanka	South Asia	0.54	0.13	0.68	0.69	0.17	0.45	0.45
Indonesia	East Asia	0.57	0.14	0.69	0.73	0.06	0.42	0.45
Nepal	South Asia	0.55	0.12	0.69	0.65	0.1	0.37	0.44
Philippines	East Asia	0.53	0.18	0.69	0.71	0.11	0.36	0.44
Ghana	Africa	0.47	0.13	0.65	0.64	0.07	0.62	0.43
Zambia	Africa	0.45	0.05	0.66	0.48	0.15	0.67	0.41
Pakistan	South Asia	0.55	0.08	0.68	0.65	0.06	0.15	0.41
Togo	Africa	0.43	0.07	0.71	0.52	0.1	0.51	0.39
Burkina Faso	Africa	0.4	0.04	0.69	0.62	0.14	0.48	0.39
Ethiopia	Africa	0.44	0.04	0.63	0.53	0.09	0.28	0.38
Mozambique	Africa	0.41	0.05	0.65	0.47	0.13	0.56	0.37
Bangladesh	South Asia	0.44	0.11	0.68	0.59	0.05	0.35	0.37
Madagascar	Africa	0.41	0.04	0.7	0.39	0.09	0.55	0.36
Rwanda	Africa	0.42	0.07	0.66	0.37	0.14	0.53	0.36

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PYTHON KPI DASHBOARD

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Developed KPIs to highlight the overall performance of the Global Resilience Index and establish a foundation for deeper analysis.

```
...  ============================================================
       GLOBAL RESILIENCE INDEX — KPI SUMMARY
     ============================================================
       Average Resilience Score          0.5479
       Average Risk Score                0.677
       Food Dependency Rate              nan%
       Food Vulnerability Score          0.876
       Highest Survival Score            0.6795
       Lowest Stability Score            0.1517
       Most Resilient Country            Austria
       High Risk Countries               29
       Stable Countries                  71
       Most Balanced Country             Germany
       Most Improved Country             China
       Lowest Risk Country               Pakistan
       Strongest Domain                  Food Security
       Weakest Domain                    Healthcare
       Top Region                        Oceania
       Lowest Region                     South Asia
       Undernourishment Rate             8.0065%
     ============================================================
```

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Double Exposure: Healthcare & Undernourishment

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A22 - Double Exposure: Healthcare & Undernourishment

	Country Name	Region	Healthcare Score	Avg Undernourishment %	Exposure Type
5	Bangladesh	South Asia	0.05	14.56	Double Risk
68	Pakistan	South Asia	0.06	15.49	Double Risk
39	Indonesia	East Asia	0.06	11.54	Double Risk
33	Ghana	Africa	0.07	8.80	Single / No Risk
51	Madagascar	Africa	0.09	33.87	Double Risk
28	Ethiopia	Africa	0.09	24.83	Double Risk
61	Nepal	South Asia	0.10	10.92	Double Risk
87	Togo	Africa	0.10	20.47	Double Risk
72	Philippines	East Asia	0.11	12.37	Double Risk
86	Thailand	East Asia	0.12	9.60	Single / No Risk
38	India	South Asia	0.12	15.00	Double Risk
59	Mozambique	Africa	0.13	28.50	Double Risk
58	Morocco	Africa	0.13	4.83	Single / No Risk
67	Oman	Middle East	0.14	7.84	Single / No Risk
52	Malaysia	East Asia	0.14	3.17	Single / No Risk
12	Burkina Faso	Africa	0.14	15.47	Double Risk
77	Rwanda	Africa	0.14	34.33	Double Risk
13	Cambodia	East Asia	0.14	10.23	Double Risk
96	Viet Nam	East Asia	0.14	10.43	Double Risk
97	Zambia	Africa	0.15	41.13	Double Risk

Highlights countries facing both weak healthcare systems and high undernourishment rates, identifying those with the highest combined vulnerability.

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TOP DECADE IMPROVEMENTS

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A13 - Decade Scores & Improvement

	Country Name	Region	2000s	2010s	2020s	Decade_Improvement
10	Bosnia and Herzegovina	Europe	0.45	0.53	0.63	0.18
17	China	East Asia	0.46	0.55	0.64	0.18
58	Mongolia	East Asia	0.44	0.50	0.61	0.17
97	Uzbekistan	Central Asia	0.43	0.51	0.60	0.17
98	Viet Nam	East Asia	0.46	0.54	0.62	0.16
0	Albania	Europe	0.47	0.55	0.63	0.16
14	Cambodia	East Asia	0.42	0.45	0.57	0.15
93	United Arab Emirates	Middle East	0.53	0.61	0.68	0.15
32	Georgia	Central Asia	0.49	0.57	0.63	0.14
57	Moldova	Europe	0.48	0.58	0.62	0.14
66	North Macedonia	Europe	0.52	0.57	0.65	0.13
46	Korea, Rep.	East Asia	0.61	0.68	0.74	0.13
1	Armenia	Central Asia	0.47	0.55	0.60	0.13
88	Thailand	East Asia	0.47	0.51	0.59	0.13
76	Romania	Europe	0.55	0.61	0.67	0.12
73	Philippines	East Asia	0.41	0.44	0.53	0.12
79	Saudi Arabia	Middle East	0.52	0.57	0.64	0.12
21	Cyprus	Europe	0.54	0.59	0.66	0.12
18	Colombia	South America	0.48	0.53	0.60	0.12
2	Australia	Oceania	0.63	0.68	0.74	0.12

Highlights the countries that achieved the greatest improvement in their Composite Score from the 2000s to the 2020s, demonstrating long-term resilience progress.

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MACHINE LEARNING PIPLINE

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```
100 In-Filter Countries
        |
        ▼
OLS Forecasting
        |
        ▼
2030 Forecast


163 Out-of-Filter Countries
        |
        ▼
Machine Learning (Linear Regression)
        |
        ▼
Estimate Domain Scores
        |
        ▼
OLS Forecasting
        |
        ▼
2030 Forecast
```

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FORECASTING MODEL

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```
Forecast rows (Country × Domain): 600
Expected: up to 600 (100 × 6)

Sample forecast rows:
Country Code Country Name Region                 Domain  Base_2030  Optimistic_2030  Pessimistic_2030
         ALB      Albania Europe       Climate & Energy   0.730576         0.740583          0.720568
         ALB      Albania Europe Digital Infrastructure   0.840727         0.868371          0.813084
         ALB      Albania Europe     Economic Fragility   0.725811         0.743920          0.707702
         ALB      Albania Europe          Food Security   0.838294         0.908667          0.767920
         ALB      Albania Europe             Healthcare   0.325711         0.351780          0.299641
         ALB      Albania Europe    Political Stability   0.754361         0.790109          0.718614
```

Component	Details
Model	OLS Linear Regression
Purpose	Forecast resilience trends to 2030
Input	Domain Scores (2000–2023)
Outputs	• Base Forecast • Optimistic Scenario • Pessimistic Scenario

```
=== BACKTEST METRICS (2021–2023 Holdout) ===
                Domain   RMSE    MAE  Coverage_%  N_Predictions
    Economic Fragility 0.0635 0.0502        20.0            300
            Healthcare 0.0653 0.0452        21.3            300
   Political Stability 0.0660 0.0491        37.7            300
         Food Security 0.0731 0.0489        49.3            300
      Climate & Energy 0.0920 0.0672        10.0            300
Digital Infrastructure 0.0978 0.0796        15.7            300

Saved: backtest_results.csv, regression_params.csv
```

Component	Details
Objective	Validate the forecasting model before predicting 2030
Training Period	2000–2020
Testing Period	2021–2023 (Holdout Set)
Evaluation Metrics	RMSE • MAE • Prediction Interval Coverage
Result	✅ Model achieved acceptable accuracy and was used for the 2030 forecast

MACHINE LEARNING MODELS

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```
  Climate & Energy            | LR RMSE=0.0271  DT RMSE=0.0307
  Digital Infrastructure      | LR RMSE=0.0302  DT RMSE=0.0444
  Economic Fragility          | LR RMSE=0.0275  DT RMSE=0.0326
  Food Security               | LR RMSE=0.0457  DT RMSE=0.0532
  Healthcare                  | LR RMSE=0.0298  DT RMSE=0.0597
  Political Stability         | LR RMSE=0.0000  DT RMSE=0.0175

All domain models trained.
```

Point	Detail
Model Structure	One model for each Domain
Primary Model	Linear Regression
Benchmark Model	Decision Tree
Validation Method	5-Fold Cross Validation
Model Selection	Select model with lowest RMSE

MACHINE LEARNING MODELS

```
Out-of-filter estimated domain score rows: 19,874
Out-of-filter countries covered          : 163
Year range                               : 2000 – 2023
```

Step	Process	Output
1	Feature Alignment	Raw indicators matched to trained model input format
2	Missing Value Handling	Gaps filled using domain-level averages
3	Model Prediction	Resilience scores estimated for unseen countries
4	Domain Score Generation	Individual scores produced per domain per country
5	Final Output	Complete resilience score dataset for 2000–2023 used in ranking and forecasting

```
Out-of-filter forecast rows      : 868
Out-of-filter countries forecasted: 163

Saved: out_of_filter_scores.csv
```

Element	Details
Method	OLS Trend Regression on historical resilience scores
Target Year	2030
Countries Forecasted	163 countries
Domains Forecasted	All 6 domains → aggregated into Composite Resilience Index
Scenario: Base	Continues current historical trend as-is
Scenario: Optimistic	Assumes accelerated improvement (+10% trend boost)
Scenario: Pessimistic	Assumes slowdown or decline (−10% trend adjustment)
Final Output	Composite Resilience Index per country per scenario for 2030

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DASHBOARD USING PYTHON

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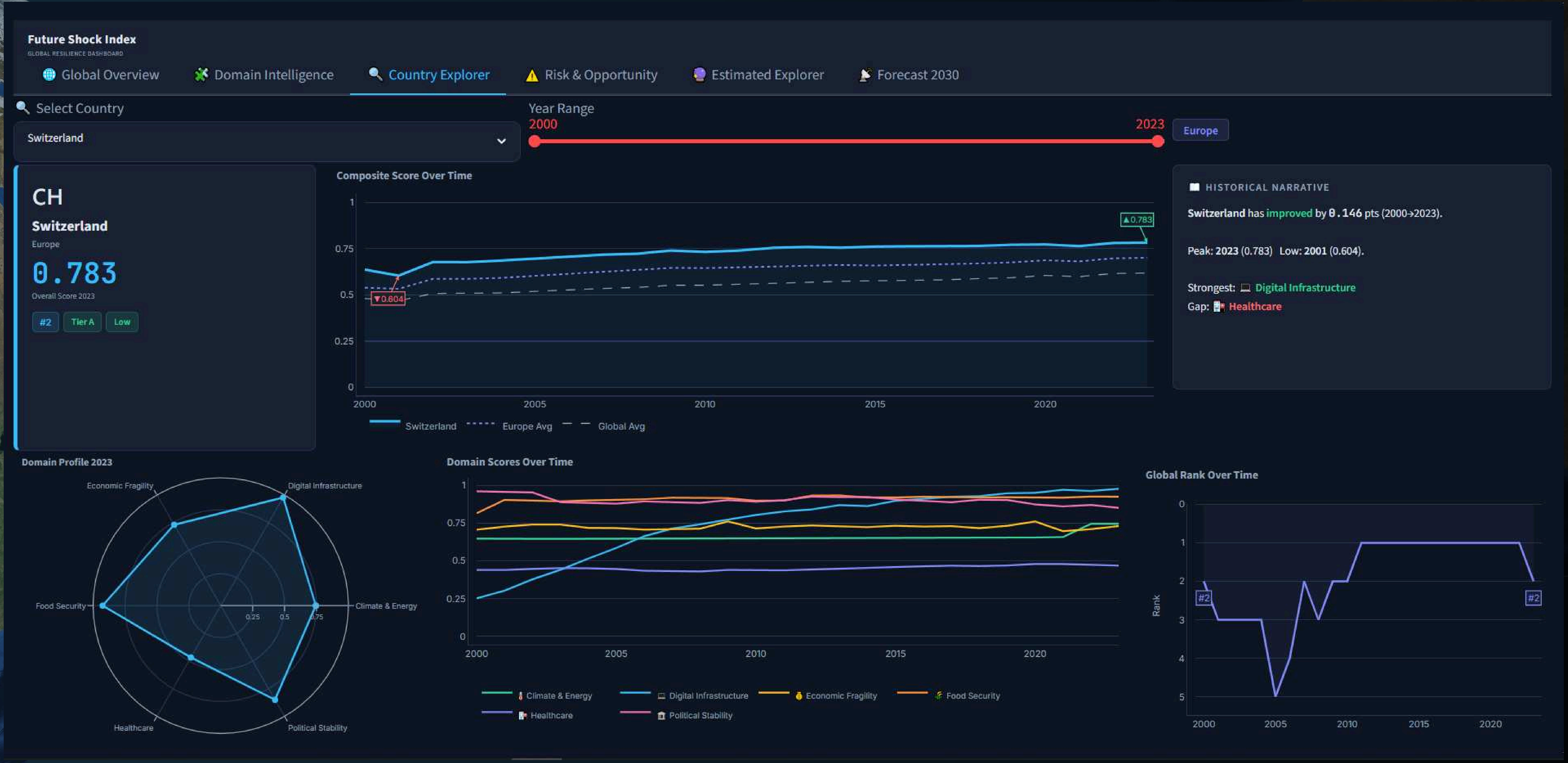
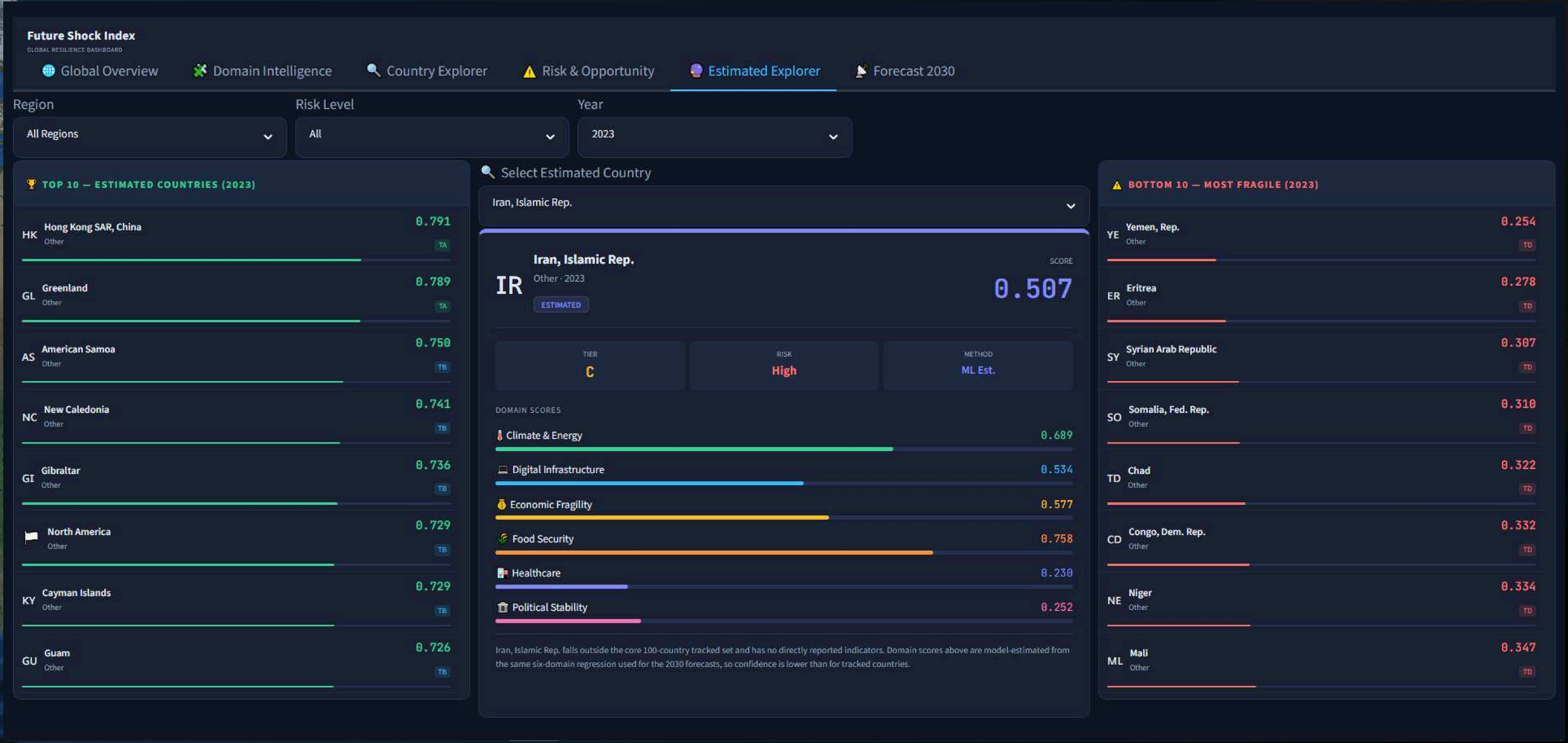
DASHBOARD USING PYTHON



ML PREDICTION DASHBOARD



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ML RESILIENCE FORECAST DASHBOARD

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Future Shock Index
GLOBAL RESILIENCE DASHBOARD

Global Overview | Domain Intelligence | Country Explorer | Risk & Opportunity | Estimated Explorer | Forecast 2030

Select Country for 2030 Forecast

Ukraine

Ukraine
UA Europe
ACTUAL

2030 BASE
0.601

RANK	TIER	RISK	CONF.
#139	B	Medium	High

DOMAIN FORECASTS (BASE 2030)

Domain	Score
Climate & Energy	0.679
Digital Infrastructure	0.818
Economic Fragility	0.783
Food Security	0.770
Healthcare	0.438
Political Stability	0.117

SCENARIO ANALYSIS — 2030

Optimistic 0.646
RANK ~#107 TIER B RISK Medium

Base 0.601
RANK ~#139 TIER B RISK Medium

Pessimistic 0.556
RANK ~#161 TIER C RISK Medium

Optimistic–Pessimistic spread of 0.089 points to **high sensitivity to policy decisions**.

EXECUTIVE RECOMMENDATION

STRENGTHS

Ukraine's strongest projected domain is **Digital Infrastructure** (Base: 0.818), indicating structural resilience capacity through 2030.

AREAS TO IMPROVE

Political Stability (Base: 0.117) is the critical gap. Targeted investment here could materially improve overall rank.

PRIORITY ACTIONS

Scenario spread of 0.089 indicates **high sensitivity to policy decisions**. Prioritize stabilizing lowest-scoring domains and leverage **Digital Infrastructure** as a platform.

EXPECTED 2030 OUTCOME

Under the base scenario, **Ukraine** projects Tier **B** by 2030 at score 0.601 (Rank #139 globally), maintaining a resilient standing in **Europe**.

DASHBOARD USING POWER BI

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DASHBOARD USING POWER BI

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DASHBOARD USING TABLEAU

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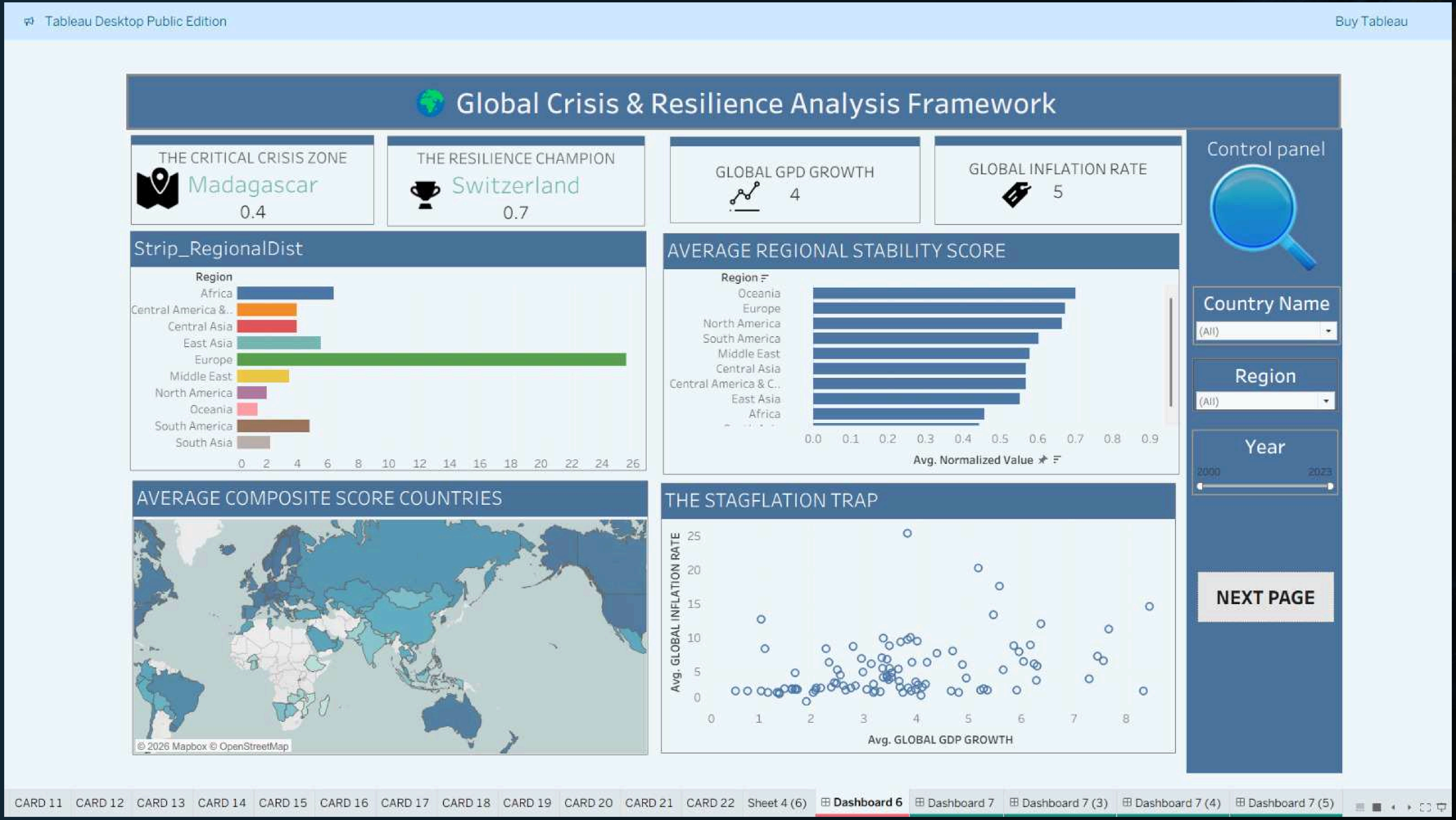
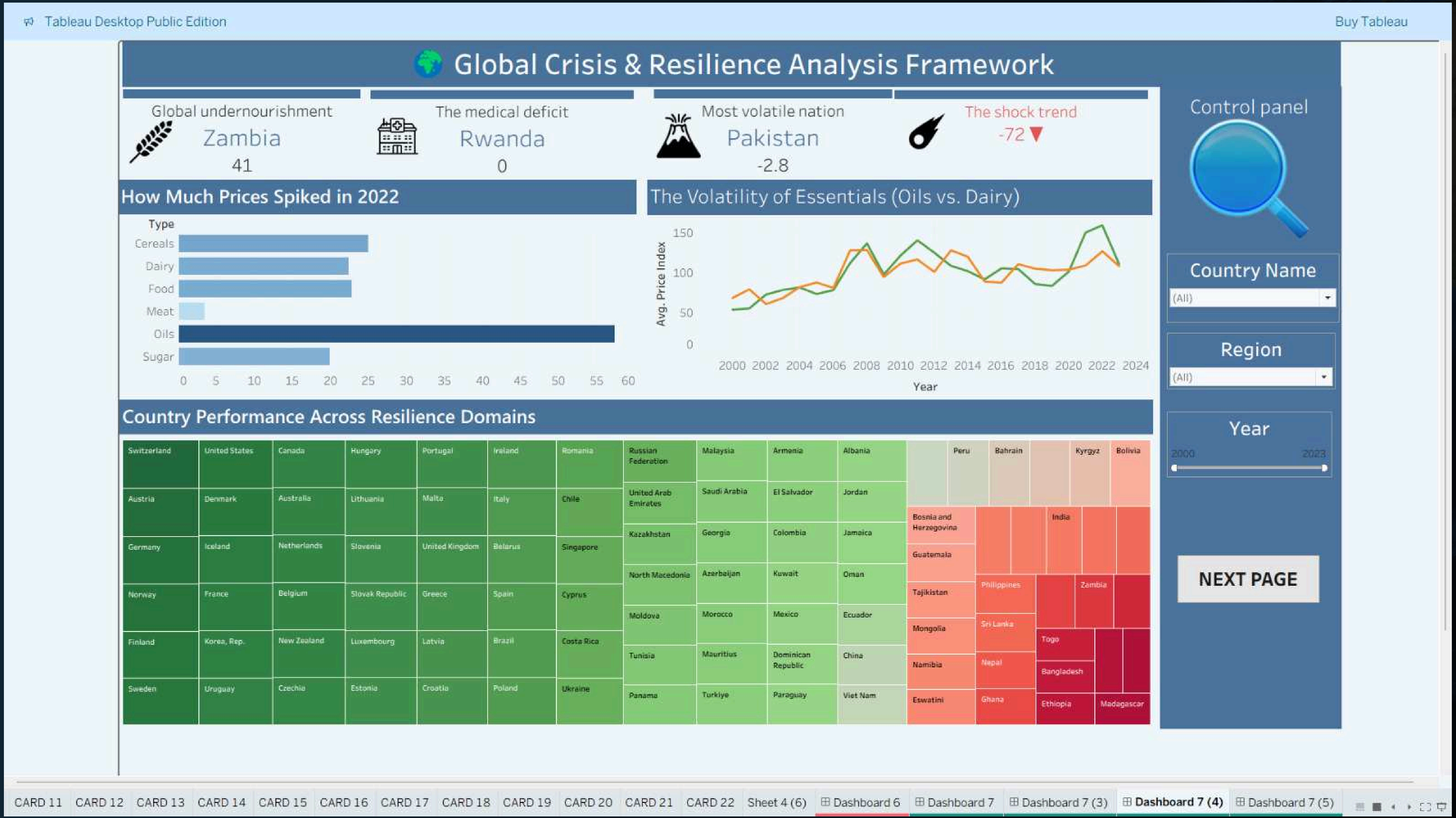


DASHBOARD USING TABLEAU



FUTURE SHOCK INDEX

STATISTICAL ANALYSIS AND KEY INSIGHTS

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18% OF DATA POINTS WERE IDENTIFIED AS OUTLIERS

Most outliers were found in GDP and Inflation, highlighting the need for careful trend interpretation.

Healthcare Remains the Weakest Domain

Healthcare recorded the lowest resilience score globally, highlighting it as the highest priority for future investment.

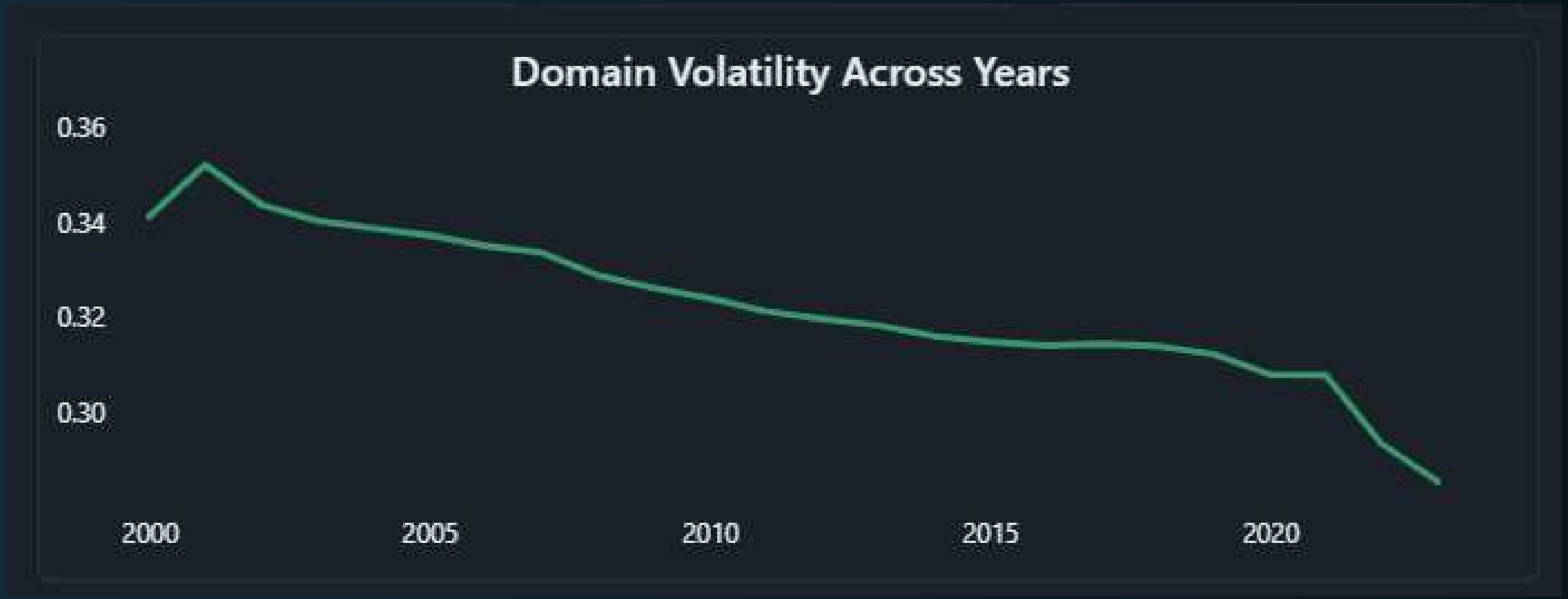
COVID-19 RECOVERY FOLLOWED BY A NEW GLOBAL SUPPLY SHOCK

Global GDP recovered after COVID-19, but rising energy and food prices in 2022–2023 created a new economic shock.

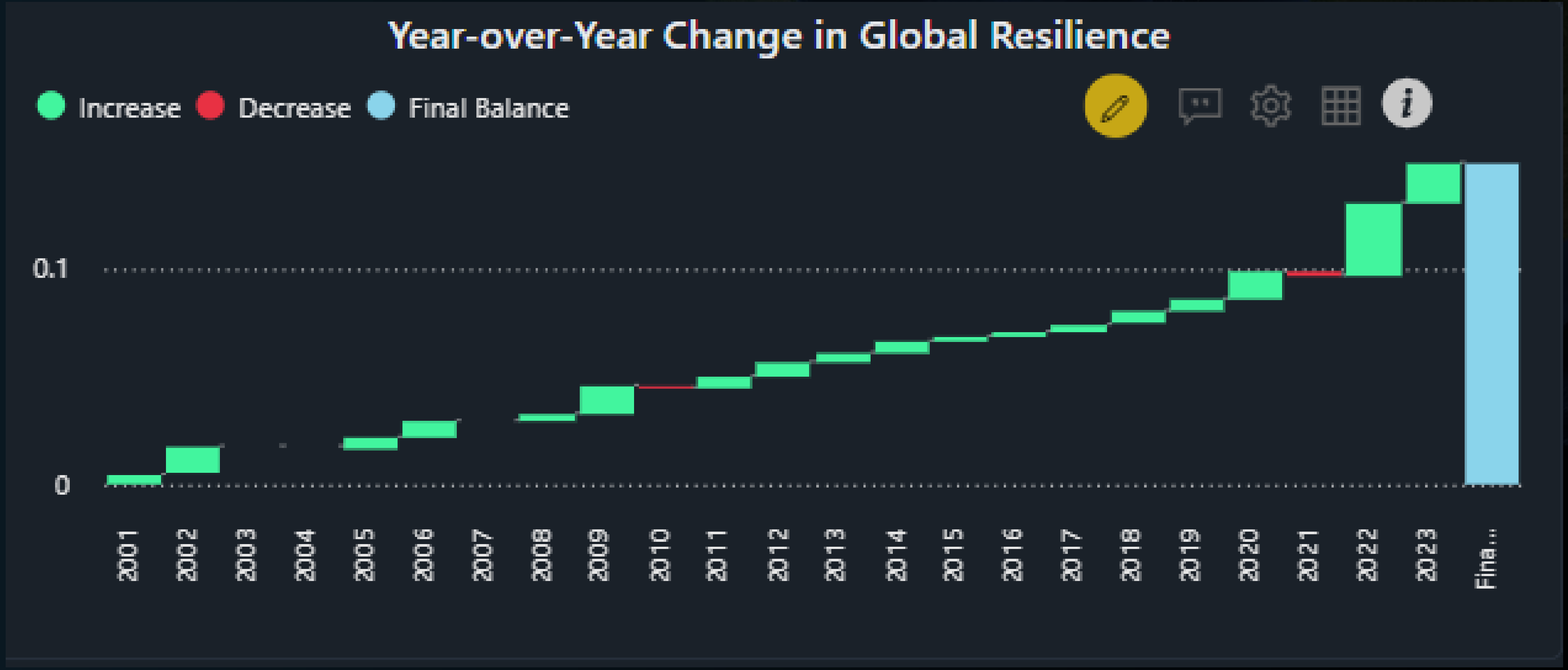
Europe Shows the Most Consistent Resilience

European countries exhibit the most consistent resilience scores, reflecting strong institutional stability.

STATISTICAL ANALYSIS AND KEY INSIGHTS



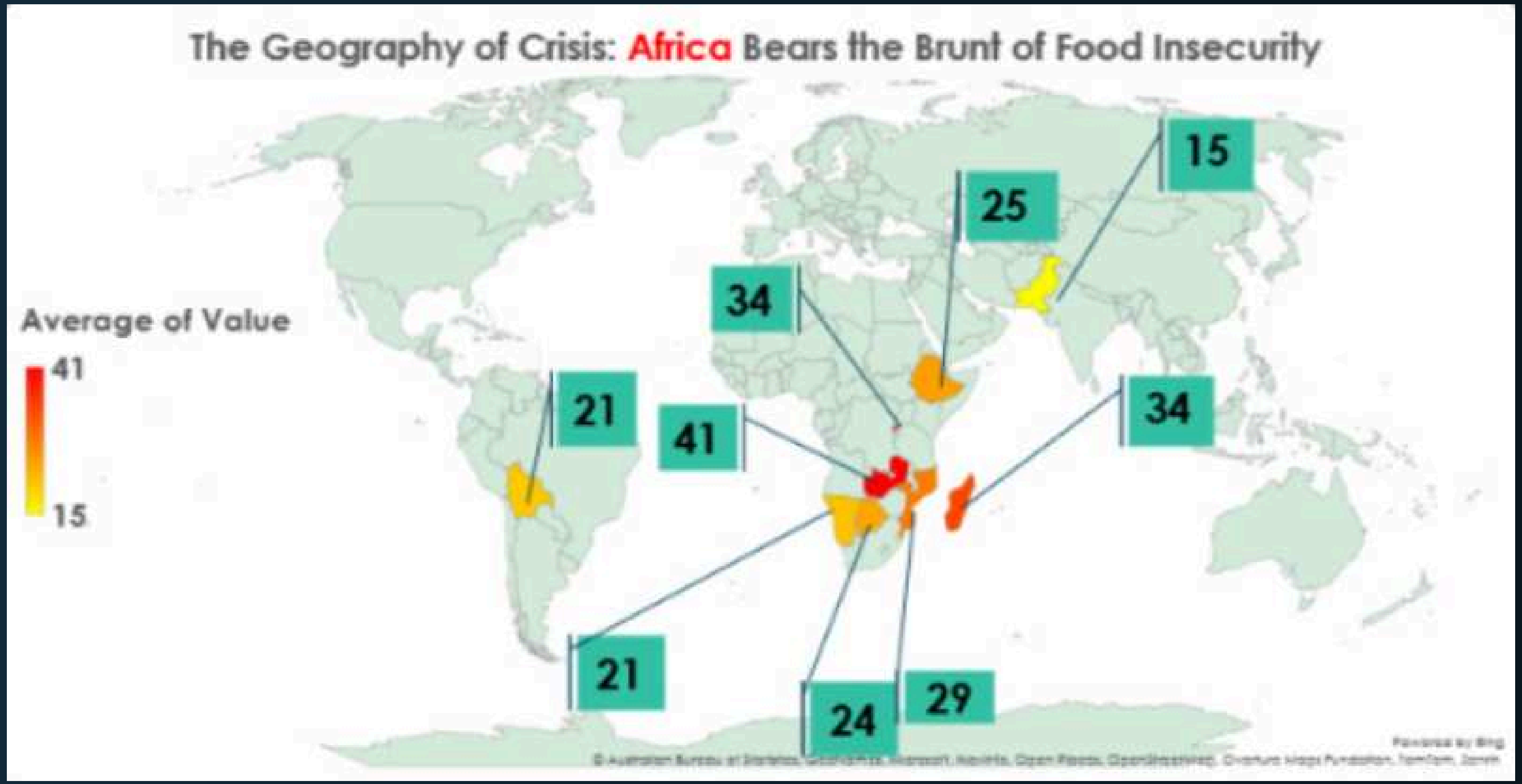
- Domain volatility declined from 2000 to 2023, indicating more stable resilience over time.
- The largest decline in domain volatility occurred after 2020.



- Steady improvement across most years.
- 2022 recorded the largest positive contribution.
- Only minor declines occurred in a few years.
- Overall trend shows sustained long-term growth.

STATISTICAL ANALYSIS AND KEY INSIGHTS

⇒ Edible oils experienced the largest price increase, while meat remained the most stable.



⇒ Food insecurity is highest in Africa, highlighting significant regional disparities.

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KEY INSIGHTS AND FINDINGS

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Digital Infrastructure is Rising Fastest

Global internet access expanded significantly, strengthening overall resilience despite declining political stability.

Electricity Access is Bimodal

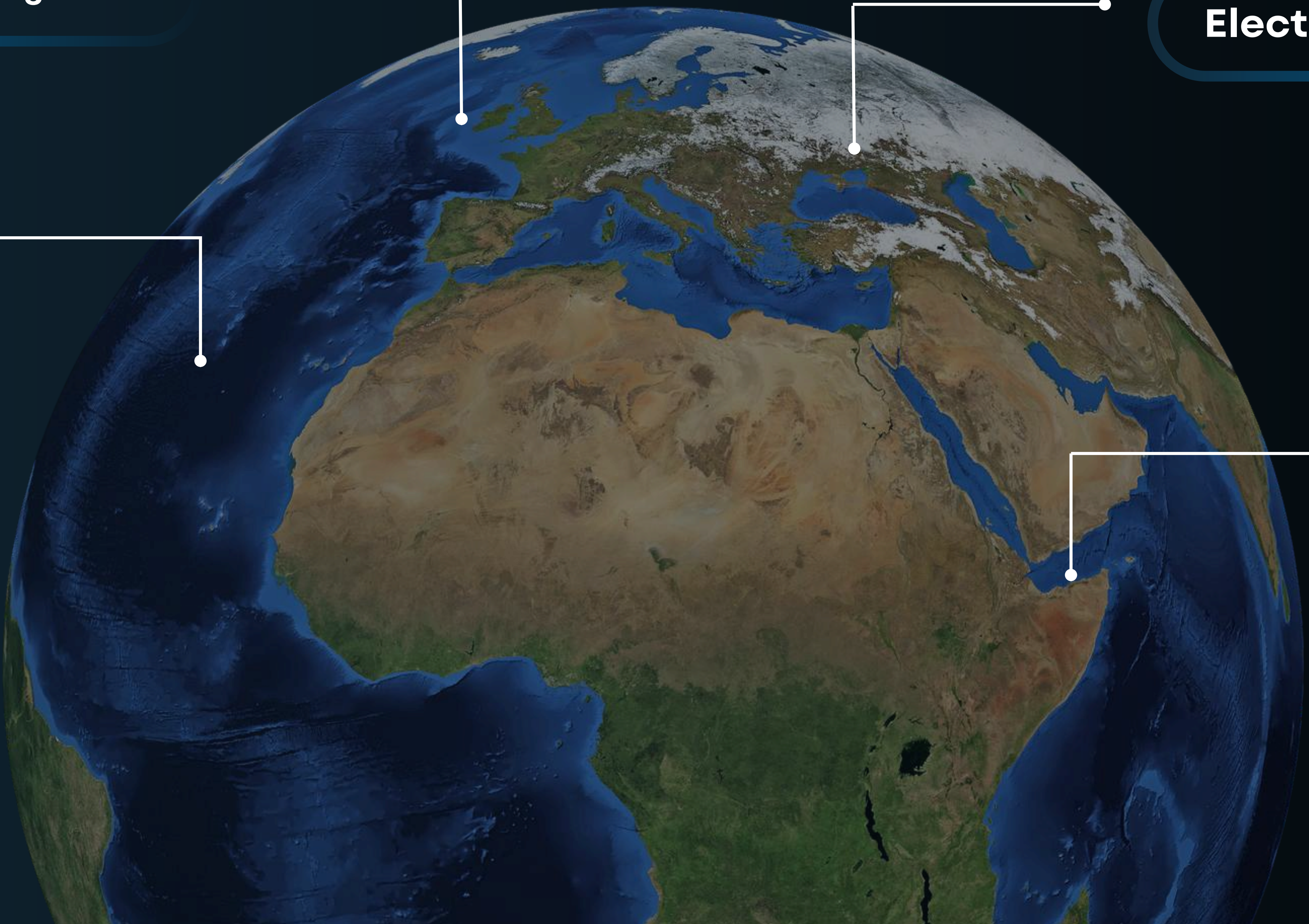
60% of countries have near-universal internet access, while 25% remain below 50%, leaving only 15% in between.

Political Stability is Declining

Political Stability is the only domain to decline consistently since 2010, with large differences across countries.

Regional Gap is Structural

Europe maintained a higher average resilience score than Africa (0.74 vs. 0.41) across all decades.



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CONCLUSION AND RECOMMENDATION

Strengthen Healthcare Systems
Increase health spending to at least 5% of GDP to strengthen pandemic preparedness, especially in the 19 high-risk countries.

Reduce Food Import Dependency
Store enough food supplies for at least 90 days and reduce dependence on a single import source.

Expand Sustainable Energy
Expand electricity access through renewable energy, especially in the 25% of countries with limited power access.

Implement Resilience Monitoring
Develop a real-time dashboard that alerts when resilience scores decline or food prices exceed critical levels.

Q&A


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THANK YOU.